

ZipDepth as a Backbone for YOLO26n — Short Report

Lab entry task · Amirmahdi · September 2026 · KITTI 2D detection, Google Colab (Tesla T4)

Question. Can features learned for monocular depth estimation replace a detection backbone? YOLO26n (**Run A**) is compared with the same detector whose backbone (layers 0–10) is replaced by the depth-pretrained ZipDepth-base encoder (**Run B**). The encoder's outputs s_2 / s_3 / s_4 have the same strides (8 / 16 / 32) as YOLO's P_3 / P_4 / P_5 , so 1×1 Conv-BN-SiLU adapters (96/192/384 → 128/128/256 channels) connect them to the **unchanged** YOLO26 neck and head, which start from the same COCO weights as Run A. Both models are fine-tuned with identical settings (50 epochs, 640 px, batch 16, SGD lr 0.01, seed 0; 5,985 train / 1,496 val images) and evaluated back to back. Twelve pre-training checks passed, including exact reproduction of the original encoder outputs and an overfit test.

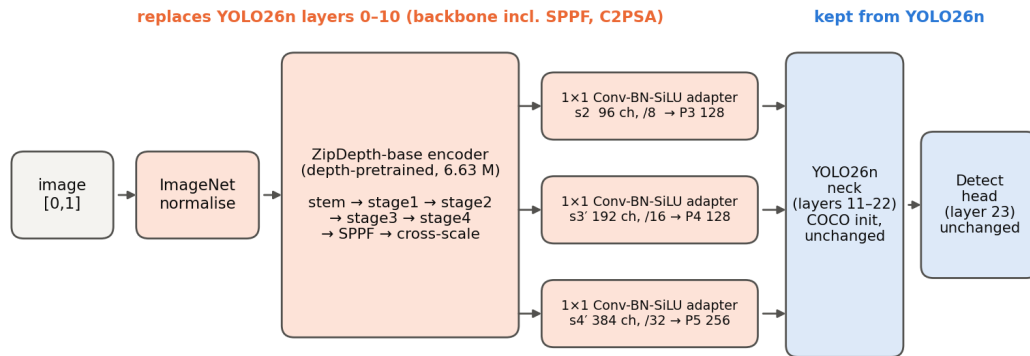


Figure 1. Run B: ZipDepth encoder + adapters replace YOLO26n layers 0–10; neck and head are kept.

Results

Metric (val: 1,496 images, 8,128 objects)	A: YOLO26n	B: ZipDepth	B / A
mAP50-95	0.588	0.472	0.80
mAP50	0.849	0.731	0.86
AP50-95 small cars (< 25 px)	0.550	0.470	0.85
Parameters (fused)	2.38 M	7.14 M	3.00×
GFLOPs @ 640	5.32	17.82	3.35×
Latency, batch 1, FP16 (T4)	12.3 ms	11.8 ms	0.96×
Throughput, batch 32, FP16 (T4)	513 img/s	318 img/s	0.62×
Peak GPU memory: training / inference b32	2,423 / 466 MiB	2,829 / 786 MiB	1.17× / 1.69×

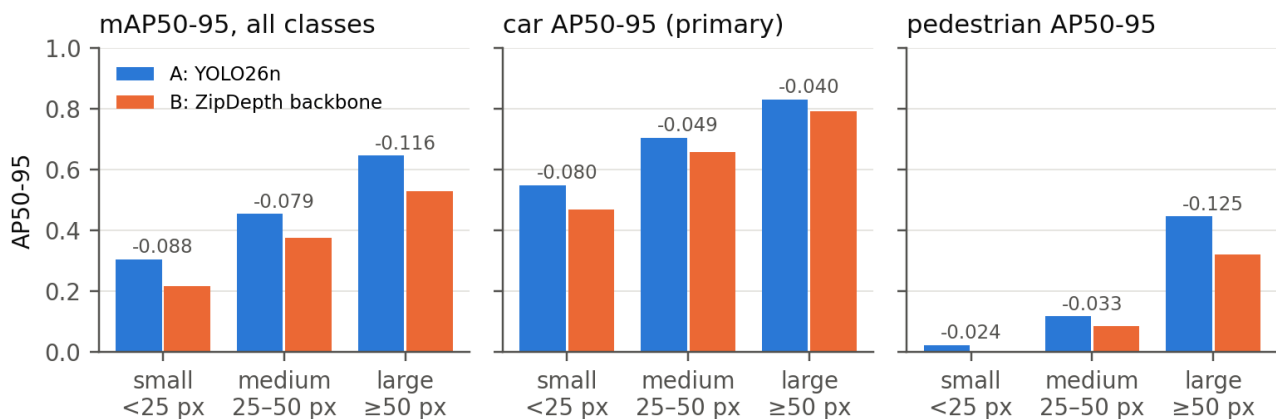


Figure 2. AP50-95 by ground-truth box height (small < 25 px, medium 25–50 px, large ≥ 50 px). Labels give B – A.

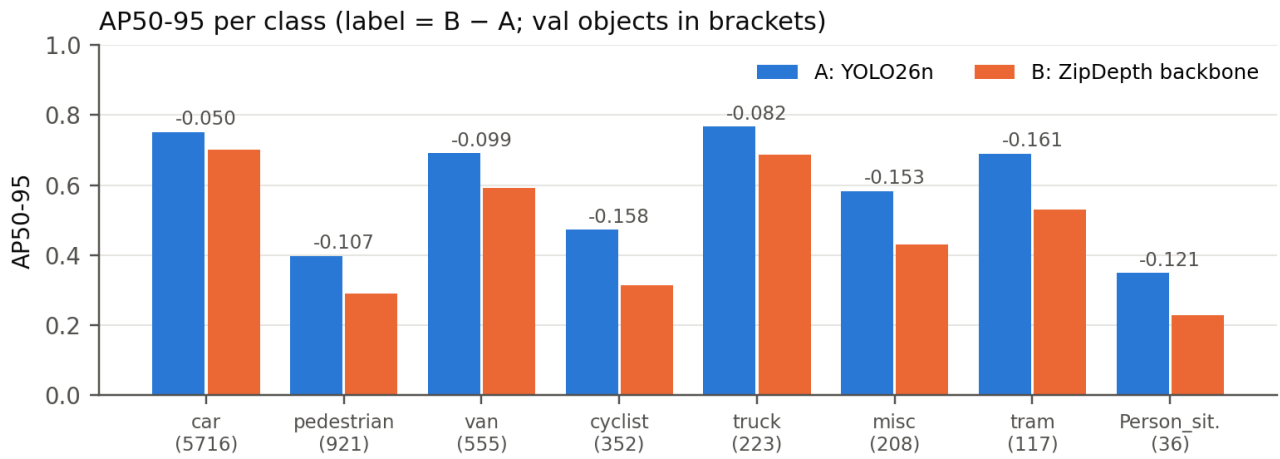


Figure 3. AP50-95 per class (val objects in brackets).

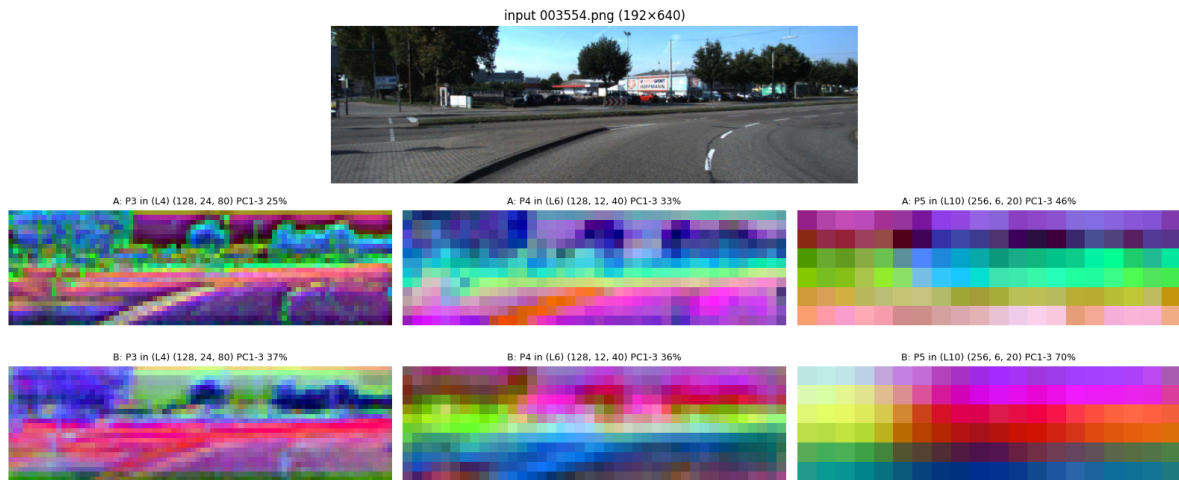


Figure 4. PCA-RGB of the backbone outputs handed to the neck (P3 / P4 / P5) after fine-tuning; row 1 Run A, row 2 Run B. Titles give the variance explained by three components.

Findings

- **Less accurate:** -0.116 mAP50-95, lower on all 8 classes (cars -7% , person classes -27 to -35%). Errors are mainly missed objects (recall $0.762 \rightarrow 0.679$), not class confusion.
- **Depth hypothesis not supported:** the relative loss is largest for small, distant objects (small cars -15% vs -5% for large cars).
- **More expensive:** $3\times$ parameters and $0.62\times$ throughput. Batch-1 latency is unchanged on the T4 only because it is launch-bound; training memory $+17\%$.
- **Feature maps:** Run B's P5 is a smooth layout / depth gradient (70 % of variance in three components vs 46 % for A) and its P3 has less local detail; the fine-tuned neck makes both models' neck outputs similar but cannot restore that detail.
- **Caveats:** Run B had not converged at 50 epochs; single seed; the validation split also selected best.pt. Run C (random encoder), needed to isolate the value of depth pretraining, has not been run yet.